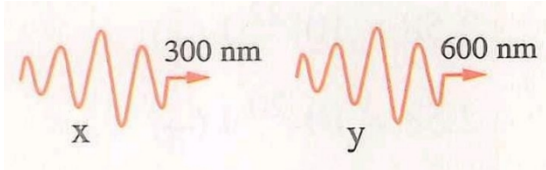


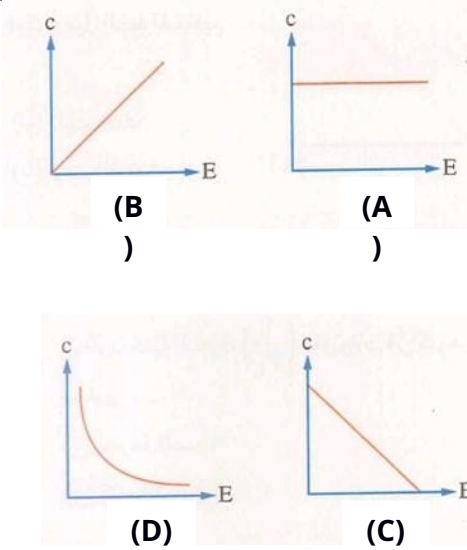
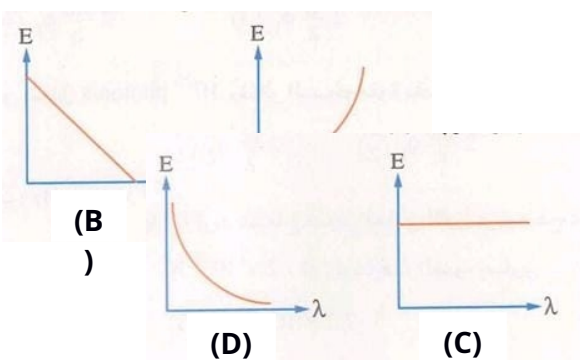
المعرفة	الكود
Compton Scattering	TPK02

1-Choose the correct answer:

#	Question	Choices	Answer
1	The ratio between the energy of the photon to the velocity of light in air is the of the photon	a. Mass b. Frequency c. Kinetic energy d. Momentum	d
2	the ratio of the photon's momentum to its mass equals	a. The velocity of light b. Planck's constant c. The photon energy d. photons are massless particles	a
3	in Compton effect, ratio between the photon energy after collision to its energy before collision is	a. Greater than one b. Equal one c. Less than one	c
4	In Compton effect, the ratio between the velocity of photon after collision to the velocity of photon before collision	a. Greater than one b. Equal one c. Less than one	b
5	In Compton effect, the ratio between the frequency of photon after collision to the frequency of photon before collision	a. Greater than one b. Equal one c. Less than one	c

6	In Compton effect, the ratio between the wavelength of photon after collision to the wavelength of photon before collision	a. Greater than one b. Equal one c. Less than one	a
7	in Compton effect, the speed of gamma rays photons after collides with a free electron is	a. Increased b. Decreased c. Equal zero d. Does not change	d
8	in Compton effect, gamma photons is scattered and its increases	a. Energy b. Speed c. Wavelength d. Momentum	c
9	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon frequency decreased and changed its direction; also the electrons velocity increased and changed its direction	a. Characteristic radiation b. Continuous radiation c. Compton effect d. Thermionic emission	c
10	The Compton effect proves	a. The particle nature of photon. b. The wave nature of photon. c. The wave nature of electron. d. The particle nature of electron .	a
	In Compton effect, when a photon with high frequency collides with free electron , which of the following will	a. Energy . b. Frequency. c. Wavelength. d. Linear momentum.	c

	increase for photon after collision		
	In Compton effect after collision , there is no decrease of	a. The wavelength of electron. b. Energy of photon. c. Photon frequency. d. Photon speed.	d
	From the properties of photon	a. Its speed equals the speed of light . b. Deflects by the electric field. c. It may be accelerated. d. All the above.	a
	At the opposite figure, the ratio between the speed of photon x and y is..... 	a. $\frac{4}{1}$ b. $\frac{2}{3}$ c. $\frac{1}{1}$ d. $\frac{1}{3}$	c
	The linear momentum for a photon is	a. $\frac{h}{\lambda}$ b. $\frac{hv}{\lambda}$ c. $\frac{hc}{\lambda}$ d. $\frac{hv}{c^2}$	a
	Photon x has P_L while photon y has $2 P_L$, so	a. The wavelength of x is double wavelength of y. b. The wavelength of x is half wavelength of y.	b

		c. The speed of x is double speed of y. d. The speed of x is half speed of y.	
	Two photon the ratio between their frequencies is 1:2, so the ratio between their energies is	a. 1:1 b. 1:2 c. 2:1 d. 4:1	b
	A monochromic light its number of photons is n and its wavelength is λ , so the total energy is	a. $\frac{hc}{n\lambda}$ b. $\frac{n\lambda}{hc}$ c. $\frac{nhc}{\lambda}$ d. $\frac{nc}{\lambda}$	c
	Which of the following graphs represents the relation between the speed of photon and its energy	 (B) (A) (D) (C)	a
	Which of the following graphs represents the relation between the energy of photon and its wavelength.....	 (B) (D) (C)	d

	Photons are electrically neutral particle and they are not deflected by the either electric or magnetic field.....	a. True b. False c. Non of the above	a
	1. According to Compton effect when a photon collides with a free electron. Then,	a. the photon loses its energy. b. the electron gains a kinetic energy equals that of the photon energy. c. the frequency of the scattered photon is less than the frequency of the incident photon. d. the electron and the photon move together on the same line.	c
	In Compton experiment , the sum of the photon energy and the electron energy before collision is the sum of their energies after collision.	a. greater than c. equal to b. less than d. no correct answer	c
	In Compton effect, the ratio between the velocity of photon after collision to the velocity of photon before collision.....	a. Greater than one b. Equal one c. Less than one	b

2- True or False

No.	Question	Answer	Correction if F
1	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, The photon's frequency decreases	T	
2	When a photon of high frequency (x- rays or gamma rays) collides with a free electron,	F	increase

	The photon's wavelength decreases		
3	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed decreases.	F	Stays the same
4	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed increases.	F	Stays the same
5	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed remains the same.	T	
6	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the electron's speed increases.	T	
7	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the electron's speed decreases.	F	
8	Classical theories can explain the change in wavelength.	F	Can't
9	The conservation of energy states that energy of the photon before and after collision is the same.	F	
10	The conservation of energy states that energy of the electron before and after collision is the same.	F	
11	The conservation of energy states that energy of the sum of energy of photon and electron before and after collision is the same.	T	

3-Solve the following:

N	Question	Answer
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0.		
1	A photon of X rays of ($\lambda = 3 \text{ \AA}$) collides a stationary electron, the electron is scattered with energy $1.1 \times 10^{-16} \text{ J}$, so the wavelength of the scattered photon is $\text{\AA}$	Photon 0 $\lambda = 3 \text{ \AA} = 3 \times 10^{-10} \text{ m}$ $E = h \frac{c}{\lambda} = 6.6 \times 10^{-16} \text{ J}$ <i>before collision</i> electron stationary $\rightarrow v = 0$ $KE = \frac{1}{2} m v^2$ $KE = 0$ <i>before</i> $KE = 1.1 \times 10^{-16} \text{ J}$ <i>after</i> $(E_p + E_e)_b = (E_p + E_e)_a$ $6.6 \times 10^{-16} + 0 = \frac{hc}{\lambda} + 1.1 \times 10^{-16}$ $5.5 \times 10^{-16} = \frac{1.99 \times 10^{-25}}{\lambda}$ $\lambda = 3.6 \times 10^{-10} \text{ m}$
2	calculate the photon frequency, mass and linear momentum if $\lambda = 380 \text{ nm}$	To calculate the photon frequency, mass, and linear momentum, we can use the formulas: - Photon frequency (ν): The frequency of a photon can be calculated using the equation: $\nu = c / \lambda$, where c is the speed of light and λ is the wavelength of the photon. - Photon mass (m): Photons are massless particles, so their mass is zero. - Linear momentum (p): The linear momentum of a photon can be calculated using the equation: $p = h / \lambda$, where h is Planck's constant and λ is the wavelength of the photon. Given that $\lambda = 380 \text{ nm}$, we can calculate the values as follows: - Photon frequency (ν): $\nu = c / \lambda = (3.0 \times 10^8 \text{ m/s}) / (380 \times 10^{-9} \text{ m}) = 7.89 \times 10^{14} \text{ Hz}$. - Photon mass ($m$): Photons are massless particles, so their mass is zero. - Linear momentum (p): $p = h / \lambda = (6.63 \times 10^{-34} \text{ J}\cdot\text{s}) / (380 \times 10^{-9} \text{ m}) = 1.75 \times 10^{-27} \text{ kg}\cdot\text{m/s}$
3	the energy of photon $3.3 \times 10^{-19} \text{ joule}$. Calculate its mass, frequency and wavelength.	a. photon is massless b. $f = E/h$ $f = 4.98 \times 10^{14} \text{ Hz}$ c. wavelength = c/f $6.02 \times 10^{-7} \text{ m}$

	From studying Compton effect, mention what happens after collision for each value of the following, then mention the reason : (a) The photon energy. (b) The photon velocity.	The energy of photon after collision is decreased. The photon velocity after collision remains constant.
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4- Mention the scientific term

No	Question	Answer
1	The total linear momentum is the same before and after collision	The conservation of momentum
2	The total energy is the same before and after collision	The conservation of energy